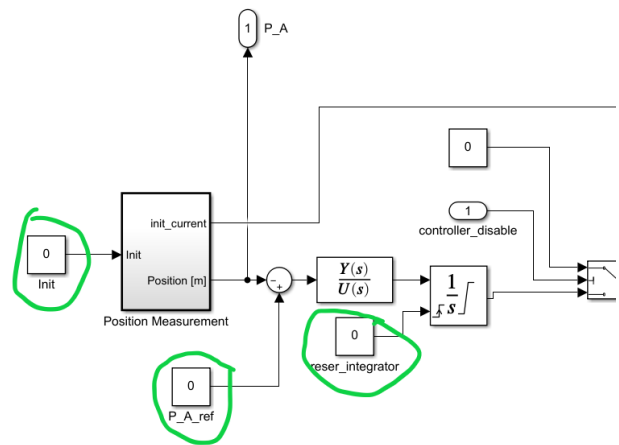
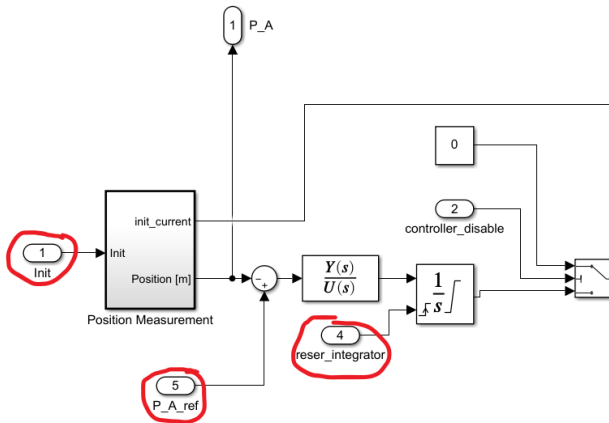


Start up manual control desk and setup

Starting up



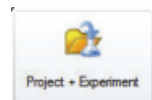
Switch on the power supplies (8), the micro box (7) and the three linear motors (4, 5, 6). Start up MATLAB and open the Simulink model, choose the RT1202! Make sure that the reference, init and reset for the integrator are constants see figures below. Labels are not controllable through ControlDesk. Make sure these constants are set at 0, otherwise the motors can start moving the moment you build the Simulink file onto the micro box. You can also have some other reference but make sure you use a switch to ensure the motor does not start moving without permission through ControlDesk. Now implement your controller and build the Simulink model. Your SDF file will be build in the file you have currently opened in MATLAB.



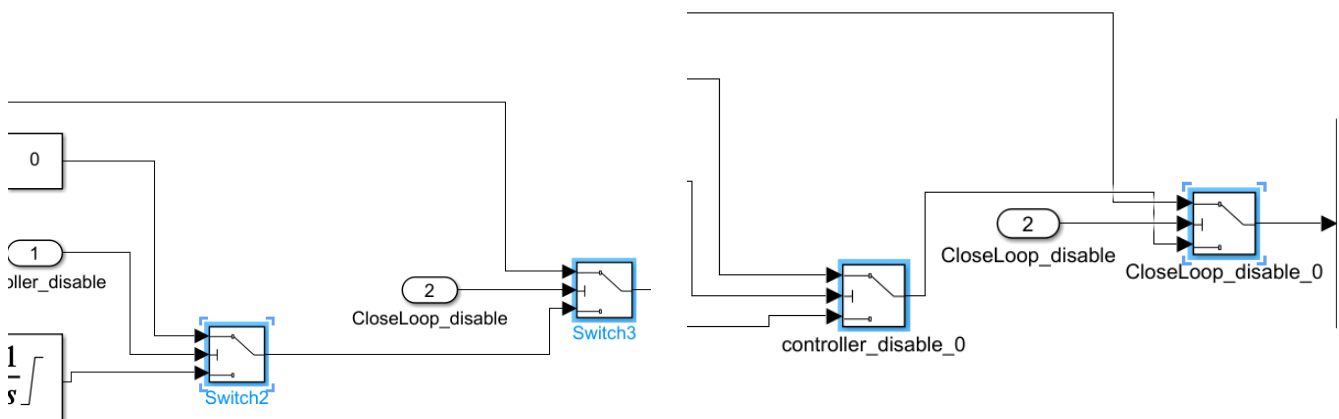
Now start ControlDesk.

Setting up control desk

When starting up control desk, make a new project and new experiment. You can put that project where you want to. It asks what device you want to use, click on the DS MICROLABBOX, it is selected as standard so DON'T change it. Next you have to select the SDF file, choose the SDF file you just computes in Starting up.

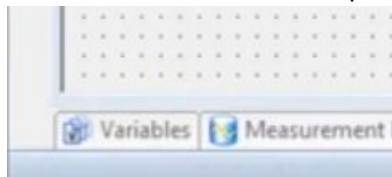


Switches



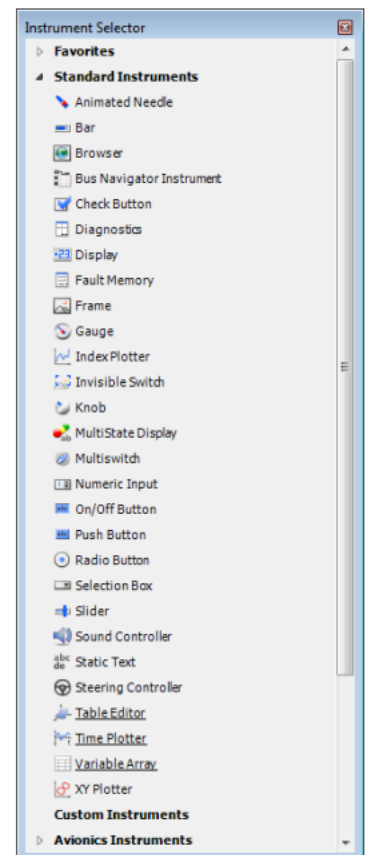
First you have to add six buttons for the switches. You can rename the switches as can be seen in figures above. This is to easily find them in control desk. Now in control desk add radio buttons. The buttons can be found on the right side under instrument selector. You have to change a few properties. So once you added these 6 radio buttons, for each one use right mouse click and select instrument properties. You will find a selection button with three dots as option. Click on those dots, and adjust the following things: change the name Button 1 into disable, change the variable of Button 1 into 0. Change the name Button 2 into enable, change the variable of Button 2 into 1.

Now you have to give variables to the buttons. We want to be able to switch the controller and closed loop on. Go to variables in the left bottom. Then click on Model Root, here you can find all the constants, variables and outputs of the Simulink model. Find your switches and click on them. You can now see two things, the output of the switch or the Threshold. Select



the Threshold and drag this one to the radio button. You can now use the switches.

Favorite	Variable Type	Cor	Variable	Block	Platform/Device
☐	 Measurement		Control_out	Model Root	Platform
☐	 Measurement		SigGenOut	Model Root	Platform
☐	 Measurement		Throttle_act_pos	Model Root	Platform



Resets

We have two reset buttons per motor, one for the integrator and one for the encoder. The Init is for the encoder and the reset is for the integrator. Here we again need 6 radio buttons as described in switches. For each button use right mouse click and select instrument properties. You will find a selection button with three dots as option. Click on those dots, and adjust the following things: change the name Button 1 into running, change the variable of Button 1 into 0. Change the name Button 2 into reset, change the variable of Button 2 into 1.

For every simulation make sure to reset both, this can be done by choosing the button reset and put it back to running.

Position measurement

We want to see the position of each encoder, therefore we need three displays. These can again be found in instrument selector. Put those in Control desk and now find the Position Measurement in the variables list. When you click on that, you get the current or the position measurement. Drag the position measurement for each encoder to a display. You can now see the position of the motors.

If the motors are in rest, the position should be approximately -0.0285. If this is not the case, reset the Init flag for the encoders.

Current2v

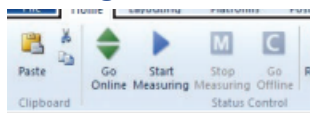
To check what we sent to the microbox we want to display the gain Current2V. We again need three displays for this. Now go to variables and find Outputs to Amplifier. In there, find the variable Current2V, drag these for every motor to the display. You can now see what you sent to the motors.

If the Current2V stays -10 or 10 even though you change the reference, try to reset the integrator by using the radio button made in resets.

Reference

To adjust the reference depending in what you designed in Simulink you can make a block for this to. If you want to just use a constant reference, you have to add a Numeric Input form Instrumental selector. Now find the constants with the corresponding names and drag them to the Numeric Input blocks. Because the encoders 0 point is at 30 mm, you can choose a reference between -30mm and 30mm. Be careful, the input is in Meters so make sure you use $30 \cdot 10^{-3}$.

Starting Control Desk



If you want to start the setup, choose the button Go online in figure.... When online, first disable all the switches, and then enable. Give a reference and now the setup should work, most of the time this is not the case.

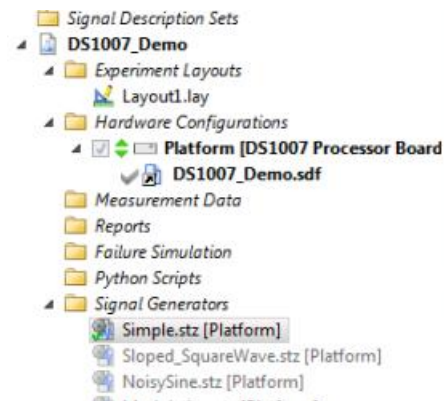
Red lights on the motors

If this is the case, switch on and off the motors with the switches connected to them (black simple round switches). Wait for a few seconds and turn them on again.

If this does not work and the lights stay red, try to switch off the complete power and the motors, NOT THE MICROBOX!!! After a few seconds turn them on again.

If this still does not work, switch off the micro box and the power. After some time restart the power, they should now be green. Now also switch on the micro box. You now have to rebuild your Simulink model!

Reload SDF file with same control desk setup



Change in Simulink whatever you want and build your Simulink model. Again the SDF file will be uploaded where you are in your MATLAB directory. Go now back to control desk and your experiment. On your left side you have a tab project, open this. You can find your platform under hardware configurations. Right mouse click on that and click on load new variables. You can now upload your new SDF file in there.